

Extrauterine Pregnancy 10.

Special-Site Ectopics: Site-Specific Logic

Companion reading handout for simultaneous listening.

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Transcript

Extrauterine Pregnancy audio course. Episode 10. Special-Site Ectopics: Site-Specific Logic.

Special-site ectopic pregnancy is not a collection of rare curiosities. It is a reminder that the treatment plan is written by the implantation bed. A tubal ampullary ectopic, an interstitial ectopic, a cervical ectopic, an ovarian ectopic, an abdominal ectopic, and a heterotopic pregnancy do not fail in the same way, bleed from the same structures, or tolerate the same interventions.

The way to listen to this lecture is to ask the same question at every site: what tissue has trophoblast invaded, and what happens when that tissue is stretched, separated, or treated? If the tissue is tube, rupture and hemoperitoneum dominate. If it is cervix, poor contractility dominates. If it is abdominal peritoneum or bowel, placental separation dominates. If it is interstitial tube in myometrium, delayed rupture and vascular bleeding dominate. The names are easier to remember when each name is attached to a tissue behavior.

Interstitial pregnancy is a tubal ectopic implanted in the interstitial segment: the proximal portion of the fallopian tube that traverses the uterine myometrium. ISUOG describes this segment as narrow, about 0.7 millimeters wide and 1 to 2 centimeters long, and surrounded by myometrium. That myometrium allows more expansion than the distal tube, so diagnosis may be delayed. But the surrounding uterine wall is vascular. When rupture occurs, bleeding can be catastrophic.

Repeat the interstitial anatomy as a picture. A narrow tube segment runs through thick uterine muscle. The muscle gives the pregnancy room to expand longer than the ampulla might. But that same location places rupture inside a vascular uterine environment. The reason interstitial pregnancy can rupture later and bleed more severely is the same reason: surrounding myometrium and uterine blood supply.

The diagnostic problem is lateral location. Not every pregnancy near the uterotubal junction is interstitial. A lateral normally sited pregnancy is still within the endometrial cavity. An interstitial ectopic is outside the endometrial cavity in the

intramural tubal segment. ISUOG and RCOG describe the ultrasound criteria as an empty uterine cavity, a gestational sac or products of conception located laterally in the interstitial intramural part of the tube, less than 5 millimeters of surrounding myometrium in all imaging planes, and the interstitial line sign. The interstitial line is an echogenic line running from the endometrial echo complex to the interstitial mass; ISUOG cites sensitivity around 80 percent and specificity around 98 percent.

The 5 millimeter criterion is a boundary of surrounding myometrium, not a random small number. Too little myometrium around a laterally placed sac supports that the sac is outside the cavity in the intramural tube rather than embedded normally in the cavity. The interstitial line sign then adds a route: an echogenic line from the endometrial echo complex toward the interstitial mass. Less than 5 millimeters plus the line sign makes the lateral pregnancy anatomically legible.

This diagnosis should not be called "cornual" loosely. ISUOG reserves cornual pregnancy for pregnancy in a rudimentary horn, a bicornuate uterine cornu, or the upper lateral portion of a septate uterus. Farren argues that confusing terminology causes clinical errors because angular, interstitial, cornual, and rudimentary horn pregnancies have different risks and management. The word should direct anatomy, not blur it.

Management of interstitial pregnancy depends on stability, size, hCG, viability, and surgical access. ISUOG states that surgical management is the main option and that conservative approaches such as cornuostomy, laparoscopy, and fertility-preserving repair are increasingly used in experienced centers. Cornual wedge resection disrupts uterine architecture and may increase later uterine rupture risk; ISUOG cites a reported subsequent rupture or dehiscence incidence of 30 percent after cornual resection in one study. Expectant or medical management is reserved for hemodynamically stable patients without immediate rupture risk, with stringent follow-up. In one conservative series, local methotrexate, systemic methotrexate, and expectant management had reported success rates of 91, 80, and 71 percent, but numbers were small and 5 of 35 conservatively managed patients ultimately needed surgery, all with hCG above 9000.

Cervical pregnancy is implantation below the internal os. The critical distinction is implantation rather than passage. Miscarriage tissue can descend through the cervix and appear low; cervical ectopic trophoblast is implanted in cervical tissue. Novak notes that cervical ectopic pregnancy may not be suspected until suction curettage for presumed incomplete abortion produces hemorrhage. Farren explains the mechanism: the cervical stroma lacks the contractile hemostatic capacity of the uterine corpus, so separation of trophoblast or placenta can cause severe bleeding.

This is one of the easiest special-site distinctions to remember if you keep the verb clear. Miscarriage tissue is passing. Cervical ectopic pregnancy is implanted. Passing tissue can move and usually lacks a trophoblastic blood supply at the cervix.

Implanted trophoblast is fixed and vascular. The danger of curettage comes from mistaking implanted tissue for passing tissue.

The ultrasound criteria make that mechanism visible. RCOG describes an empty uterus, a barrel-shaped cervix, a gestational sac below the internal os, absent sliding sign, and blood flow around the sac on Doppler. If tissue slides with gentle pressure, miscarriage in progress becomes more likely. If the sac is fixed and vascular, cervical implantation must be considered. Management may include systemic methotrexate, local potassium chloride when fetal cardiac activity is present, suction evacuation or hysteroscopic resection, and hemostatic adjuncts such as balloon tamponade, sutures, uterine artery embolization, or, when bleeding cannot be controlled, hysterectomy.

Ovarian pregnancy is located completely or partially within ovarian parenchyma. The main diagnostic trap is the corpus luteum. A corpus luteum is expected in early pregnancy, may be cystic, and can show vascularity. The scan must ask whether the lesion is inseparable from the ovary and has ectopic pregnancy morphology rather than merely showing the physiologic ovary. ESHRE defines ovarian pregnancy by location in ovarian tissue; Novak notes that ultrasound diagnosis is difficult because the finding may resemble a corpus luteum, hemorrhagic cyst, dermoid, or tubal ectopic near the ovary.

The repetition here is simple: expected ovary is not ectopic ovary. Every early pregnancy should have a corpus luteum somewhere. It can look impressive, it can have a ring of flow, and it can be painful if hemorrhagic. Ovarian ectopic diagnosis requires the clinician to decide whether pregnancy tissue is actually in ovarian parenchyma, not merely beside or confused with the ovary.

Management is usually surgical when diagnosis is made at laparoscopy. Farren states that surgical management is recommended and usually laparoscopic, with excision of the pregnancy or wedge resection of the affected ovary. Blood loss may be higher than in ordinary tubal surgery; one series cited by Farren reported mean blood loss of 700 milliliters. Serial hCG follow-up is indicated after treatment to ensure there is no active residual trophoblast.

Abdominal pregnancy is implantation in the peritoneal cavity outside the uterus, tubes, and ovaries. It may involve the pouch of Douglas, uterovesical pouch, broad ligament, omentum, bowel, pelvic sidewall, or other peritoneal surfaces. The danger is placental attachment to tissue not designed for placental separation. In the uterus, myometrium contracts and spiral-artery physiology is part of the system. On bowel, omentum, pelvic sidewall, or broad ligament, removal of the placenta can open vessels that cannot be compressed by uterine contraction.

Early abdominal ectopic pregnancy may be removed surgically, often laparoscopically if feasible. Advanced abdominal pregnancy requires individualized planning. Farren emphasizes that surgery depends on placental site, involved organs, degree of invasion, and accessibility of the blood supply, and that MRI may help planning. If removal of the placenta is expected to cause catastrophic bleeding, the placenta may

be left in situ after cord ligation, with long-term conservative management or delayed removal once blood flow has ceased. The team may need expertise in complex vascular, bowel, bladder, and pelvic surgery.

Rudimentary horn pregnancy belongs to congenital uterine anatomy. A functional non-communicating horn in a unicornuate uterus can receive a pregnancy that is surrounded by myometrium but not connected normally to the uterine cavity. Farren describes ultrasound criteria: a single interstitial portion in the main uterine body, a gestational sac separate from the uterus and surrounded by myometrium, and a vascular pedicle joining the sac to the unicornuate uterus. If diagnosed before pregnancy, excision of a functional horn can be considered to prevent this dangerous implantation. If pregnancy is already present, management is excision of the rudimentary horn and associated tube when feasible, with urgent bleeding control if rupture has occurred.

Intramural pregnancy is different again. It is implantation within the myometrium, above the internal os, separate from the interstitial tube and separate from the cesarean scar or cervix. Farren notes that it may mimic cystic fibroid or adenomyotic cyst and that Doppler assessment is essential because trophoblast should generate increased flow. Partial intramural pregnancies may sometimes be treated transcervically under continuous ultrasound guidance; complete intramural pregnancies often require abdominal, frequently laparoscopic, evacuation and repair. Expectant management may be appropriate only when hCG is declining, and biochemical and ultrasound resolution can take months.

Heterotopic pregnancy changes the diagnostic rules because a normally sited pregnancy and an ectopic pregnancy coexist. hCG no longer helps in the usual way because the intrauterine pregnancy contributes the hormone. Seeing an intrauterine pregnancy is therefore not an exclusion test, especially after assisted reproduction, ovulation induction, or severe symptoms. Methotrexate is not appropriate when a viable normally sited pregnancy is continuing. Management may involve surgery for the ectopic component or local agents such as potassium chloride or hyperosmolar glucose in selected cases where preserving the intrauterine pregnancy is the goal.

This is the place to repeat the course's location discipline one more time. Heterotopic pregnancy proves that "I found a pregnancy in the uterus" and "I excluded ectopic pregnancy" are different statements. The uterus can contain a real normally sited pregnancy while the tube also contains an ectopic pregnancy. Once a viable normally sited pregnancy is present, methotrexate is no longer a routine option because it would threaten the pregnancy the patient may wish to continue.

The practical way to handle special-site ectopics is to ask the same set of questions every time. Where is the gestational sac relative to the endometrial cavity, cervix, tubes, ovary, myometrium, scar, and peritoneum? What tissue is the trophoblast invading? What vessel bed will bleed if it separates? Can that tissue contract or be compressed? Is methotrexate treating the dominant problem, or is the dominant

problem anatomical hemorrhage risk? Can the patient be safely followed if conservative management is chosen?

When those questions are answered, the rare diagnoses become less mysterious. Each one is an ectopic pregnancy, but the site determines the danger, and the danger determines the plan.